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Challenges in Partially Automated Driving and a Proposed Solution

Case Study

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Cognition in Complex Situation

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1 Introduction

This case study examines how partial automation features, such as Drive Pilot in the Mercedes-Benz S-Class and EQS, can alter the traditional responsibilities of a human driver. While systems like Drive Pilot reduce physical workload and handle many driving tasks, they also introduce new challenges for drivers. Specifically, I discuss three main challenges regarding topics I learned in the "Cognition in Complex Situations" seminar. I then describe each challenge by explaining why it arises, what the specific problems are, and how these issues manifest in real-world scenarios. Next, I focus on solutions for divided attention and distraction. I will discuss these topics based on theories I learned in the "Cognition in Complex Situations" seminar, along with insights from related research. This subject is particularly interesting to me because I am currently conducting an experiment related to trust calibration in autonomous vehicles.

2 Challenges

2.1 Maintain situation awareness and trust

Human Machine Collaboration

2.1.1 What leads to the challenge?

While the Drive Pilot system handles the driving, drivers can remove their hands from the steering wheel and their eyes from the road to engage in various other activities such as operating the infotainment system, talking with other passengers, eating, or drinking. Even though using mobile phones is prohibited by law, drivers may still engage in these activities. Based on research conducted in Brazil, it was found that brand reputation and driver trust in autonomous driving technologies are strongly correlated (Rodrigues und Ponchio, 2020). In other words, among experienced drivers, there is a substantial increase in trust and acceptance of SAE Level 3 ADS after their first exposure to an autonomous driving system, suggesting that over time, drivers tend to trust the system beyond its actual capabilities (Schwindt-Drews et al., 2024; Foppen, 2025). Another key consideration is that most people know how advanced technology in their cars is integrated with software that continually learns and evolves over time. As a result, it can be difficult for drivers and the system to define clear boundaries in their roles, especially when emotions come into play. This situation may blur driver expectations and degrade situational awareness.

When it comes to situational awareness, it simply means knowing what's going on in the environment (Endsley, 2000). It plays a vital role in driving, even with partial automation - because when the Drive Pilot makes a takeover request, the driver must have situational awareness to respond and handle the situation in a timely manner. Research by Walch et al. (Walch et al., 2017) presents four key requirements that help foster effective engagement between the driver and the autonomous system:

- Shared situation representation: both parties maintain a common understanding of the current situation.
- Mutual predictability: the ability to anticipate each other’s future actions.
- Directability: the capacity to influence the other party’s activities as conditions and priorities change.
- Calibrated trust: ensuring that human trust in automation accurately reflects its capabilities.

From this scenario, it is clear that the lack of these four requirements leads to significant challenges.

2.1.2 What is the problem?

Overtrust in automated systems that exceed their capabilities can lead to misuse (J. D. Lee und See, 2004). As discussed above, this trust issue can reduce drivers’ attention to the driving task—especially when effective collaboration with the Drive Pilot autonomous system is critical—and can foster complacency during takeover requests. Drivers may take longer to react and might lack sufficient situational awareness to handle the situation properly, as they become mentally disengaged. This poor takeover performance can negatively affect how road situations are managed.

On the other hand, the collaboration between the driver and the Drive Pilot, which relies on maintaining a shared understanding of the driving environment, can suffer because the driver is less vigilant. Moreover, the driver’s ability to manage the car during takeover requests is diminished due to a lack of situational awareness.

2.1.3 What exactly does the challenge look like?

In this challenge, the driver, the Drive Pilot system, and the dynamic driving environment are all involved. This challenge carries serious consequences if the driver cannot react to takeover requests promptly and lacks the situational awareness needed to respond effectively, which may increase the risk of accidents or other hazardous situations. Even though the system has some countermeasures—such as parking the car in the middle of the road and activating hazard lights—there remains a high risk of ending up in a dangerous situation. Furthermore, encountering such circumstances may lead drivers and other individuals to lose trust in the system (Kraus et al., 2020).

By addressing this challenge, several positive outcomes can be achieved. For instance, user experience in autonomous driving can improve by maintaining a certain level of situational awareness while the Drive Pilot is in charge, and by defining boundaries for the driver’s and the Drive Pilot’s roles—at least to some extent—to ensure comfort, safety, and efficiency. Finally, drivers should also be educated about the exact capabilities of the Drive Pilot autonomous system.

2.2 Rapid Handover and Cognitive Load

MultiTasking and Task Switching

2.2.1 What leads to the challenge?

Since the Drive Pilot operates only under limited conditions, drivers may need to resume manual control unexpectedly based on environmental conditions. Because there is significant uncertainty in the driving environment, events can occur without warning, making time a critical factor in such situations. In the context of driving, task switching refers to drivers shifting their attention between driving and non-driving tasks (Shi und Bengler, 2022). As (Xu et al., 2024) highlights, based on analyses of various studies involving both manual driving and conditional automated driving-transitioning from a cognitive task to a driving task results in higher reaction times and lower driving stability than when drivers are not engaged in any cognitive tasks.

2.2.2 What is the problem?

In theory, this is referred to as a task-switching cost. In terms of cognitive load, while the Drive Pilot is active, drivers often engage in other tasks that degrade their situational awareness (Wickens, 2008). In this situation, when a driver must take control of the car during a takeover request, they have to quickly gather a large amount of information from the external environment to maintain situational awareness, which significantly increases cognitive load. Therefore, moving from a relaxed mode to a driving mode in a short period of time is a major challenge.

2.2.3 What exactly does the challenge look like?

In this challenge, the driver, the Drive Pilot system, and the dynamic driving environment are all involved. In particular, traffic conditions, weather conditions, and unexpected scenarios that may occur while driving can trigger this challenge. It can lead to various consequences, such as collisions or sudden stops with hazard warnings, especially if the driver's response to a takeover request is delayed or if they cannot quickly process the high cognitive load needed to regain situational awareness.

Because the Drive Pilot operates only under limited conditions, these scenarios might happen repeatedly, causing drivers to feel frustrated or anxious, which ultimately lowers user experience and satisfaction. Addressing and reducing this challenge can significantly improve user experience—not just through color-coded or on-screen messages but also by implementing clearer voice commands and smoother communication methods, thereby alleviating current user experience constraints. In addition, it is important to educate drivers about the existing limitations and role boundaries of both the driver and the Drive Pilot, as previously discussed.

2.3 Divided attention and distraction

Attention in driving-distraction/ Inattention

2.3.1 What leads to the challenge?

Driving a car is a multitasking situation in which the driver must pay attention to various aspects and process information simultaneously. Therefore, divided attention is a critical factor in this complex multitasking performance (J. Lee et al., 2003; Lengenfelder et al., 2002). In conditional partial automation (Level 3) driving, while the Drive Pilot is active, the driver can pursue other activities within certain limits.

Because of this—as discussed in the first challenge, drivers have a tendency to over-trust the car’s automation (Schwindt-Drews et al., 2024). Similarly, according to the MiRA concept, spare capacity allows drivers additional time to focus on other tasks apart from driving (Kircher und Ahlstrom, 2017). Reducing mental workload in automated vehicle drivers has the potential to affect drivers’ attention (Young und Stanton, 2002), and research claims that, in the event of an automation failure, drivers cannot regain control effectively (Stanton et al., 2001).

Another contributing factor to this challenge is the technology-rich environment. Mercedes-Benz S-Class and EQS, as its high-end flagship products, are equipped with the latest tech equipment, including a large infotainment system, an interactive driving cluster, built-in Apple/Android smartphone connectivity, a mood-change ambient lighting system, high-end surround sound, massage seats, and more. All of these can distract the driver from the road and present a significant factor leading to accidents (Stutts et al., 2005).

2.3.2 What is the problem?

When it comes to the core issue of the problem, this divided attention—caused by multitasking and inattention—leads the driver to fail to recognize what is happening in the environment. In other words, the driver ends up with a lack of situational awareness. This becomes especially critical when the driver gets a takeover request from the Drive Pilot system during a critical situation.

2.3.3 What exactly does the challenge look like?

The problem involves various factors, such as driver behavior (e.g., overtrust/confidence, brand loyalty, attitudes while driving), the Drive Pilot automation system, and the environment (e.g., traffic conditions, weather, technological constraints, unexpected situations). This challenge can have serious consequences, as research shows that 60–80 percentage of accidents involve inattention and distraction (Phase, 2006), indicating a high likelihood of car accidents. In addition, it can create a domino effect: a car accident or near-collision may reduce trust in the Drive Pilot, even if the underlying cause is driver inattentiveness or distraction. In some cases, the car may even be stopped unnecessarily or activate hazard warnings without cause, potentially leading to traffic violations.

By effectively addressing this challenge, it is possible to keep the driver’s attention while the Drive Pilot is active, at least to some extent. One approach is to monitor the driver’s actions using sensors, a feature some manufacturers already use particularly to detect whether the driver is asleep. With the advent of AI and the growing use of built-in selfie cameras, it becomes increasingly feasible to monitor and analyze driver behavior and then respond intelligently. This approach will help enhance the overall user experience. From my perspective, allowing the driver to become distracted will eventually be part of fully automated autonomous vehicles in the future.

3 Solution

3.1 Solution for Divided Attention

3.1.1 Recap of the challenge

This challenge arises when the driver relies more on the Drive Pilot partial automation (SAE Level 3 ADS), thereby redirecting attention to secondary tasks such as infotainment systems, phone use, or conversation with fellow passengers. As a result, the driver’s situational awareness of the road environment may degrade, reducing readiness to respond to unforeseen events (Stutts et al., 2005, Dingus et al., 2006). The outcome is a heightened crash or near-crash risk, especially if the automated system requires an urgent handover.

3.1.2 Proposed Solution

Driver Monitoring System In the drive monitoring system, I propose integrating eye and face tracking using a camera to measure the driver’s eye gaze direction, facial expressions, and head orientation. This technology has already been implemented by various OEM providers (e.g., Tobii¹ and Smart Eye²) and is continuously being developed because of its importance. Recent research shows that using an IR camera installed near the steering wheel, two detection techniques—one for inattention and one for drowsiness—can be employed (Kim et al., 2023). Inattention is detected when the driver does not face forward, while drowsiness is identified by monitoring for closed eyes. Additionally, the system should be able to recognize facial emotions, as these are a critical factor; research shows that anger and happiness are associated with more driving errors than fear or a neutral emotional state (Rodrigues und Ponchio, 2020, Roidl et al., 2014). By analyzing these factors, the system can provide warnings in various ways—for example, through sound alerts, message alerts on the infotainment system and driver cluster, changes in the ambient lighting color, seat vibrations (if the massage seat option is available), and voice commands. This solution can guarantee a minimum level of road attention even if the driver engages in secondary activities by continuously evaluating eye gaze and head orientation. Ultimately, it reinforces the concept of information sampling (Kircher und Ahlstrom, 2017).

¹<https://www.tobii.com/products/automotive/tobii-autosense>

²<https://www.smarteye.se/solutions/automotive/driver-monitoring-system/>

Restrict infotainment and other things that lead to distraction Based on various conditions and feedback from the driver monitoring system, non-driving-related activities can be restricted when they lead to driver inattention (Regan und Strayer, 2014). However, these restrictions should only be applied under certain conditions, not universally. Artificial intelligence can also be used to manage this more effectively, thereby maximizing user experience without compromising safety. For example, the infotainment system can be temporarily blocked or its functions limited, and the sound system can also be restricted. These temporary restrictions are well aligned with EU guidelines (»Commission recommendation of 22 December 2006 on safe and efficient in-vehicle information and communication systems: update of the European Statement of Principles on human machine interface«, 2007).

4 Conclusion

In this case study, we have seen that while partial automation, such as Drive Pilot, can reduce drivers' physical workload, it also carries risks when drivers become overly distracted. My proposed solution, real-time driver monitoring coupled with smart infotainment restrictions, aims to preserve a minimum level of attention so that drivers remain prepared for sudden takeover requests. By preventing excessive multitasking and maintaining a safety margin, we address some of the core dangers of divided attention and inattention. However, this approach still has limitations. Some drivers may find restricted infotainment features frustrating, and the sensors might misread certain head or eye positions. Nevertheless, implementing even a basic driver-monitoring system can help reduce risky behavior and support drivers' situational awareness. Through consistent refinement, user education, and careful balancing of technology and human factors, we can move closer to safer and more comfortable automated driving experiences.

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Statement of Originality

I hereby confirm that I have written the accompanying thesis by myself, without contributions from any sources other than those cited in the text and acknowledgements.



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